IP FABRIC

CHROME EXTENSION

User Guide

How to use the overlays the extension adds to IP Fabric: embedded Global Search (top-left) and the LLM Interactive Session (top-right), the CVE Security Status Evaluate control on the header row, and the Diff Dashboard — a snapshot-to-snapshot column comparison opened from a header-row button.

Audience IP Fabric users **Version** 5.1.3

Once installed and configured, the extension adds a single 48 px bar to the top of every page on demo2.eu.ipfabric.io. The bar hosts two independent tools (Global Search and the LLM Interactive Session), and the header row additionally carries the CVE Security Status control and the Diff Dashboard button. Results from the LLM Interactive Session and the CVE Security Status evaluation carry clickable links straight back into the IP Fabric UI — see “Clickable links back into the IP Fabric UI” below. This guide walks through what each one does and how to use it day to day.

1 The toolbar at a glance

Both tools live side by side in one bar — no overlap and no white gap. The left half is Global Search; the right half is the LLM Interactive Session. Each side has its own input, its own results dropdown that opens below the bar, and its own gear (⚙) shortcut to Options.

You can show or hide each overlay independently in Options → Display:

- **Show Global Search** (top-left) — on / off
- **Show LLM Interactive Session** (top-right) — on / off
- **Show CVE Security Status** — on / off (the label and Evaluate button on the header row)
- **Show Diff Dashboard** — on / off (the header-row button that opens the snapshot comparison panel; independent of the bar)

If you hide all three tools

The bar collapses to a small "IP Fabric Toolbar" restore group — an IP logo, a label, a Show options button, and a gear — so you always have a way back into Options to re-enable a feature. While any one of the three (Global Search, LLM Interactive Session, or CVE Security Status) is shown, that feature's own gear is reachable, so this restore group stays hidden.

Top-left tool — Global Search

Find every IP Fabric table that contains a value — by scanning the data itself, or by asking an LLM to investigate.

Global Search lives at the top-left of the bar. Type a value — a hostname, an IP, a VLAN name, an interface, a serial number — press Enter, and the results appear in a dropdown below the bar. There are two search modes, selectable in Options.

Each match has a clickable deep link to the corresponding IP Fabric UI page plus a preview of the first matching row.

Mode A — Cycle every table (default)

When you search, the extension fans out parallel requests to the IP Fabric API for every UI table in its bundle and checks whether your term appears in any cell. Tables that contain the term are listed in the dropdown with a clickable deep link plus a one-line sample row.

- **Uses your existing session.** This mode runs in the host page context, so it relies on your IP Fabric session cookie — no extra credentials needed.
- **Live progress.** The dropdown streams a progress bar, a scanned / total counter, and matches as they're found.
- **Speed.** A typical scan takes 5–30 seconds, depending on row counts and how many tables exist on your instance.
- **All tables are reached, including Hosts.** The scanner maps each UI table to its IP Fabric API endpoint — including non-obvious ones such as Inventory → Hosts, which is served from the addressing section of the API. Once fetched, values like a Host IP address (e.g. 172.16.61.60) are matched, whether the column is a plain string, a structured / deep-link value, or an integer-encoded IP.

The results table has four columns:

Column	What it shows
	The IP Fabric table that contained a match.
	A clickable link to the corresponding IP Fabric page.
	How many rows in that table contained the term.
	A one-line preview of the first matching row.

Mode B — LLM and MCP Server

Instead of scanning tables locally, this mode sends your query to your configured LLM (Anthropic Claude by default, or any OpenAI-compatible provider), with the IP Fabric MCP server attached as a tool. The model reasons about the query, calls MCP tools to investigate — typically several round-trips — and returns a structured answer grouped into buckets:

Bucket	What it groups
	Managed networks, subnets, VRFs that mention the term.
	Hostnames, serial numbers, login IPs, vendors.
	Location / region groupings.
	Running- or startup-config snippets.
	Which inventory snapshots contain the term.

Bucket	What it groups
	Anything that doesn't fit the buckets above (e.g. a VLAN name, an interface, a VRF, a route), with the table / column it matched.

The dropdown shows one section per non-empty bucket with the matched name and a short reason. The header shows the total match count, the number of tool calls the model made, and a one-sentence summary above the sections.

This mode needs API keys

LLM and MCP Server mode requires an Anthropic API key (or an OpenAI-compatible provider) configured in Options. Per-search cost is whatever the model's input / output tokens come to — usually a few cents per query. The default Cycle every table mode needs no keys.

Reading and managing Global Search results

Depending on the response, MCP mode picks one of three renderers: an API-endpoint table (Method · Path · Summary · Tags · Score · OK, with an expand chevron for more detail and clickable paths where the API path maps to a known UI page), a generic data table built from the union of keys across rows, or a plain JSON / text fallback.

Top-right tool — LLM Interactive Session

A free-form prompt bar — ask a question in plain language and let the model investigate your deployment through the IP Fabric MCP server.

The LLM Interactive Session sits at the top-right of the bar. Rather than searching for a single value, you type a prompt — a question or instruction in plain language — and press Enter (or click Submit). Your chosen LLM works through an agentic loop against the IP Fabric MCP server and renders the answer below the bar.

What it returns

The answer is rendered as either a structured result — bucketed into networks / devices / sites / configs / snapshots / other — or as prose, depending on what the question calls for. As with Global Search's LLM mode, results render in the right-half dropdown below the bar. Wherever an item in the answer corresponds to something in the IP Fabric UI — a device, a site, a network, a snapshot — its name is rendered as a clickable link that opens that view in a new tab.

Clickable links back into the IP Fabric UI

Each item the model returns is linked back to the IP Fabric page it came from, so you can get from the answer to the underlying data in one click. A device links to the device inventory filtered to that

exact host; a site links to the site overview filtered to that site; a network links to the managed-networks table; a snapshot links to its snapshot-management page. Links open in a new browser tab and are pinned to the snapshot the answer was computed from, so what you land on is the same data the model saw.

Only items the extension can resolve with confidence are linked. If the model returns something that doesn't map cleanly onto an IP Fabric page — or if it isn't certain which page it belongs to — the item is shown as plain text rather than as a link that would land you on an empty or unrelated view. A missing link is always preferred to a wrong one.

How a session runs under the hood

1. The extension reads your LLM provider settings and IP Fabric MCP credentials from Options.
2. It opens an MCP session with the IP Fabric server and converts the available MCP tools into tool definitions for your chosen provider.
3. It sends your prompt plus those tools to the provider, asking the model to investigate.
4. The model returns text and one or more tool calls.
5. The extension runs each tool call against the IP Fabric MCP server, feeds the results back to the model, and loops — usually 2 to 4 turns.
6. When the model finishes, the extension parses the final answer and renders it in the dropdown.

Provider choice is shared

The LLM Interactive Session uses the same LLM provider settings as Global Search's LLM mode. Set your provider, model, and keys once in Options and both tools use them. Tool-calling quality depends on the chosen model — small or local models may have weaker tool use.

Header-row tool — CVE Security Status

One-click vulnerability assessment — cross-reference your deployment's devices and OS versions against published CVE sources, right from the IP Fabric header row.

A CVE Security Status label and an Evaluate button sit centered on the IP Fabric header row, just below the toolbar (on the same row as the IP FABRIC version wordmark). The control is independent of Global Search and the LLM Interactive Session, and can be shown or hidden on its own in Options → Display. Clicking Evaluate runs a vulnerability assessment of the current deployment.

What Evaluate does

7. It drives the same agentic loop the LLM Interactive Session uses — your configured LLM (Anthropic Claude by default, or any OpenAI-compatible provider) works against the IP Fabric MCP server.

8. The model enumerates the deployment's network devices and their OS / software versions from the device and OS-version inventory tables.
9. For each vendor / OS-family / version actually in use, it cross-references published, well-known public CVE and vulnerability tracking sources — NVD, the CISA KEV catalog, and the relevant vendor security advisories (Cisco PSIRT, Juniper SIRT, Arista, Palo Alto Networks, Fortinet PSIRT, and so on).
10. It maps each relevant CVE back to the specific affected device(s) and returns a structured answer.
11. It additionally cross-references device models and OS versions for configuration-specific CVEs — those whose exploitability depends on a particular feature or configuration being enabled — and reports them in a dedicated Configuration Specific section of the output.

The result renders in a dropdown below the bar in the identical format as the Global Search and LLM Interactive Session results — a one-sentence summary plus bucketed sections (impacted hosts under Devices; configuration-specific CVEs — those whose exploitability depends on a particular feature or configuration being enabled — under a dedicated Configuration Specific section; and a per-CVE / per-version breakdown with its public source under Other), and every finding is linked back into the IP Fabric UI. If no current devices are found to be affected by any published CVEs, the summary says so. The results panel opens below the CVE Security Status control (its label and the Evaluate, results-toggle and gear buttons), so it never covers them — the control stays visible and clickable while results are shown. The control carries the same two icon buttons as the Global Search and LLM Interactive Session groups, immediately to the right of Evaluate: a results-toggle button (the list-lines icon) that shows or hides the CVE results panel — it appears only when there are results to re-show, and is highlighted while the panel is visible — and a gear button that opens the Options page. As with the other tools, the results persist until you clear them (the × in the results meta row) or click Evaluate again; clicking back on the page hides the dropdown but keeps its content, and the results-toggle button (or the next Evaluate) brings it back. Results are restored after navigation or reload, and the results-toggle button reappears so you can re-open them.

Clickable links from every CVE finding

The CVE results are hyperlinked the same way the LLM Interactive Session results are, so you can go straight from a vulnerability to the kit it affects:

- **Devices — each impacted host links to the device inventory filtered to that device, so you can see it in context (model, OS version, site, interfaces) without having to search for it.**
- **Configuration Specific — each configuration-specific CVE links to the device whose configuration makes it exploitable.**

- **Other — each per-CVE / per-version entry links to the devices running the affected OS version, so you can see everything exposed to that CVE at once.**

Links open in a new tab and are pinned to the snapshot the evaluation ran against, so the page you land on reflects the same data the assessment was based on. As in the LLM Interactive Session, a finding the extension cannot map to a specific IP Fabric page with confidence is shown as plain text rather than given a link that might mislead.

This control needs API keys

Because it uses the LLM, CVE Security Status requires an Anthropic API key (or an OpenAI-compatible provider) configured in Options — the same provider settings the LLM Interactive Session and Global Search's LLM mode use. Per-evaluation cost is whatever the model's input / output tokens come to. Cross-reference accuracy depends on the chosen model's knowledge of public CVE sources; smaller or local models may be less thorough.

Header-row tool — Diff Dashboard

Compare a Technology or Inventory table column between two snapshots — see exactly what was added, removed, or changed, without leaving the page.

The Diff Dashboard is hidden by default behind a Diff Dashboard button on the IP Fabric header row. Click the button — or press Ctrl/Cmd + Shift + D — to open its panel. The button and panel automatically position themselves to avoid the Global Search bar and the CVE Security Status control, so all four overlays can be used together.

What it does

The panel knows which Technology or Inventory table you are viewing and offers that table's full set of comparable columns — as of v5.1.2 the dropdown lists every column the table exposes — read from IP Fabric's OpenAPI specification (the same catalog the IP Fabric SDK uses), shown in the table's own order and labeling (so the routes table's network column appears as “Route”) — not just a default subset. As of v5.1.0 the Diff Dashboard recognizes every current IP Fabric Technology and Inventory table, reconciled with IP Fabric's own page definitions. Pick a column, pick a second snapshot to compare against the one currently loaded, and press Compare. The extension reads both snapshots from the IP Fabric API (using your existing session) and shows, for the column you chose:

- **Only in A** — rows present in the current snapshot but not in the compared one.
- **Only in B** — rows present in the compared snapshot but gone from the current one.
- **Changed** — rows present in both, where the column's value differs (shown as before → after).
- **Unchanged** — a count of rows whose value is identical in both snapshots.

Rows are matched across snapshots by a primary key — shown in the panel as “Rows matched on …” — mirroring the IP Fabric Python SDK's compare() behavior. This key is derived automatically per table, and (as of v5.1.3) you can adjust it: click Edit match key to choose exactly which columns identify a row — for example, key the MPLS Forwarding table on its label columns for a true per-entry comparison — or Reset to auto to return to the default. The compare column is always excluded from the key. Each result row links back into the IP Fabric UI at the relevant snapshot, and the full comparison can be exported to CSV.

How matching works: Column to compare vs. Rows matched on

Column to compare and Rows matched on do two different jobs. Rows matched on is the row's identity — it decides which row in the current snapshot is the same row as which row in the snapshot you are comparing against. Column to compare is the single field whose value you are watching for changes on those matched rows. So the match key decides how rows are paired up, and the compare column decides what counts as a change once a pair has been found.

When you press Compare, the extension buckets every row in each snapshot by its match-key value. For each key that appears in both snapshots it compares the chosen column's value(s): identical means Unchanged, different means Changed. A key found in only one snapshot becomes Only in A (removed) or Only in B (added). In short, the match key defines the buckets and the compare column is what is inspected inside each shared bucket.

This is why the compare column is always excluded from the key automatically. If the field you are watching were also part of the identity, any change to it would change the row's key — so the row would look like it vanished from one snapshot and a brand-new one appeared in the other (an add plus a remove) instead of a single Changed row.

The automatic key is a sensible default for each table, but there are three situations where you will want to add or remove columns in Rows matched on:

- **The key is too coarse (rows collapse together).** Several real entries share one key value and fall into a single bucket — for example, the default key on the MPLS Forwarding table grouped all of a device's label entries into roughly ten device rows instead of the 48 actual entries. Add a distinguishing column (such as the in-label) so each entry is matched on its own.
- **A key column is unstable between snapshots.** If a column that forms part of the identity changes value, that row's key changes, so the row is reported as Only in A plus Only in B rather than Changed — even though it is the same real object. De-select the volatile column so the row stays matched and its real change surfaces as a change.
- **The key is too fine (phantom adds and removes).** If the key includes a column that splits rows you consider the same, you will see additions and removals that are not real. De-select columns to make the match coarser.

If you un-check every column there is no per-row identity: all rows fall into one bucket, and the dashboard simply reports whether the column's overall set of values changed between the two snapshots, rather than which specific row changed.

A good rule of thumb: choose the match key so that one key value corresponds to exactly one real-world thing you care about, built from columns that stay stable for that thing across snapshots — then set Column to compare to whatever attribute of that thing you want to watch. Reset to auto restores the derived default at any time, and the key also returns to auto whenever you change the compare column or open a different table.

Settings and placement

Options → Diff Dashboard has two settings: Ignore volatile fields (age, uptime) — recommended on, so a next-hop counts as changed only when the real routing data changes, not when a timer advances — and Reset Diff Dashboard position & size. You can also drag the panel by its header, resize it from the bottom-right corner, or use the Fit to space button in its header to re-dock it into the free area above the table. Show or hide the whole feature with Options → Display → Show Diff Dashboard.

No API keys needed

Like Global Search's Cycle every table mode, the Diff Dashboard uses your IP Fabric session cookie — it does not require an Anthropic API key or the IP Fabric MCP token.

2 Everyday behaviour both tools share

Opening, hiding, and clearing results

Type your query and press Enter. Results render in a dropdown below the bar — about half the screen tall — and each side has a results-toggle button to show or hide them.

- Clicking back on the page hides the dropdown but keeps its content. The toggle button (or running another search) brings it back.
- The **×** in the results meta row clears the content for that side.

Results persist across navigation

If you follow a deep link from the results into a UI page — or otherwise reload — the last results are restored. Toggle the panel open to see them again.

The search term persists too

In Global Search, the search term itself persists as you navigate between IP Fabric pages, including when you follow a deep link from the results table into a UI page. The term stays in the input until you click the small × inside the search box.

Clearing the results dropdown — via Esc, the × in the results meta row, or clicking off — does not clear the search term. Only the explicit clear-input × does.

Keyboard shortcut

Clicking the toolbar icon, or pressing Ctrl/Cmd + Shift + F, focuses the search input. Pressing Ctrl/Cmd + Shift + D shows or hides the Diff Dashboard (you can also click its header-row button). Both shortcuts can be re-bound at chrome://extensions/shortcuts.

Save or view results as an HTML report

Whenever a tool has results, its results header shows two small buttons next to the clear (×) button — available for Global Search, the LLM Interactive Session, and CVE Security Status alike:

- **View report** (the open-in-new-tab icon) opens the current results as a formatted HTML report in a new browser tab, using your system's default handling for a new tab.
- **Save report** (the download icon) saves that same report as a self-contained .html file you can keep, email, or open later in any browser.

The report is styled to the IP Fabric brand and is fully self-contained (it needs no internet connection or external files to display). It includes the tool name, the date and time it was generated, the original input that produced the results — the Global Search term, the LLM prompt, or the CVE Evaluate action — and the results themselves, laid out the same way they appear in the dropdown. The clickable IP Fabric links in the LLM Interactive Session and CVE Security Status results are preserved in the report, so the exported file stays navigable — clicking a device in a saved CVE report opens that device in IP Fabric, exactly as it does on screen (you'll need to be logged in to the instance). Generating a report never changes or clears your current results; it simply exports what is already on screen.

3 Practical limits & troubleshooting

These apply to Global Search's Cycle every table mode:

- **Time.** 5–30 seconds per search, depending on the size of your IP Fabric instance. Each table request has a 20-second safety timeout: if a table is slow to respond or doesn't answer at all, that request is skipped (counted as no match) so the scan always finishes and reaches 100% rather than stalling.
- **Coverage.** Up to 1000 rows per table. Tables with more rows than that may have matches the scanner misses.

- **Searchable columns.** The scanner fetches a table's full column set where the API allows it, so table-specific columns like vlanName or network are searched.
- **Table coverage & API paths.** As of v5.1.0 the scanner's table catalogue is reconciled directly with IP Fabric's authoritative webPages definitions, so it covers every current Technology and Inventory table (350+ API tables across inventory, technology, cloud, SD-WAN, and more). Each UI table is mapped to its exact IP Fabric API endpoint — including sections where the API path differs from the UI path (for example Inventory → Hosts is served from the addressing section, IPv6 Routes from networks/ipv6-routes, and the Load-balancing Virtual Servers sub-tables have their own endpoints). Every listed table is fetched and searched, and tables that don't exist on your IP Fabric version are skipped silently. Once a table is fetched, a value such as a Host IP address or a load-balancer Pool name is matched — including when the column is a plain string, a structured / deep-link value, or an integer-encoded IP.
- **Semantics.** A match is any cell containing the search term as a case-insensitive substring. Very generic terms (e.g. 1) will match almost every table.

If a Global Search scan returns no results

The most likely causes are: the term genuinely isn't in any scanned table; you're not logged into IP Fabric in this tab; or your IP Fabric version uses a different API pattern than the bundled one (in which case you'll see HTTP errors in DevTools and the scan will report 0 successful tables).

If the LLM features return "401 CORS requests are not allowed for this Organization"

When the provider is Anthropic Claude, this is an Anthropic organization / account policy — not an extension bug — commonly caused by the org's data-retention settings. Resolve it by using an API key from an organization without retention restrictions and with CORS allowed, or by contacting Anthropic support to enable CORS / browser access for that organization. The Test LLM button in Options surfaces this distinctly so you can tell a policy block apart from a key or connectivity problem. OpenAI-compatible and local providers avoid this entirely.